

Foundations Recap: Qubits, Superposition & Entanglement

A visual review of the three core concepts that power quantum computation — and why they matter.



The Three Pillars of Quantum Computing

Quantum computing is built on three fundamental concepts that together enable computational advantages impossible for classical machines.

Qubits

The basic unit of quantum information — capable of existing in more than one state at a time.

Superposition

A qubit can exist in a combination of states simultaneously, enabling quantum parallelism.

Entanglement

Two or more qubits become correlated so that the state of one instantly affects the other.

CORE CONCEPT

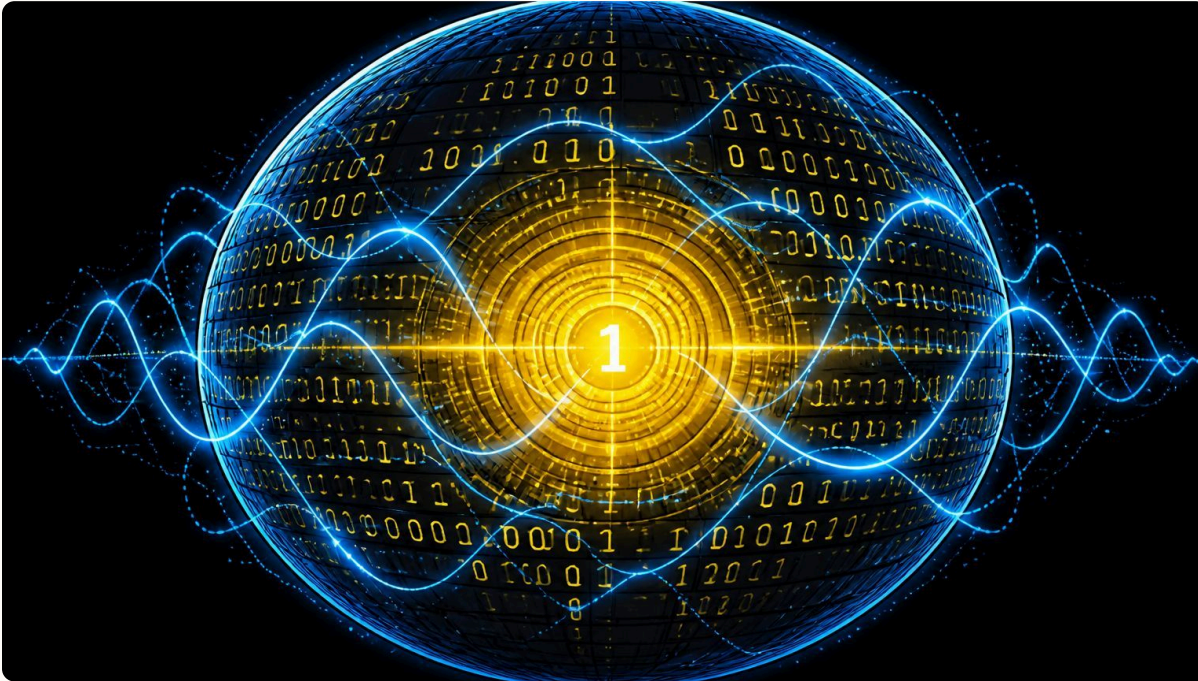
What is a Qubit?

A **qubit (quantum bit)** is the fundamental unit of quantum information. Unlike a classical bit, which is strictly 0 or 1, a qubit can exist in a combination of both states simultaneously.

The two basis states are:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Physical implementations include trapped ions, superconducting circuits, photons, and quantum dots.



Classical Bit vs. Qubit

Understanding the difference between classical and quantum information is essential to grasping why quantum computers are powerful.

Classical Bit	Qubit
Either 0 or 1	Can be 0, 1, or a combination of both
Binary, deterministic state	Quantum state described by amplitudes
No superposition	Supports superposition
No entanglement	Supports entanglement
Used in classical computers	Used in quantum computers

Superposition

Superposition allows a qubit to exist in **multiple states at the same time**. The general state of a qubit is:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

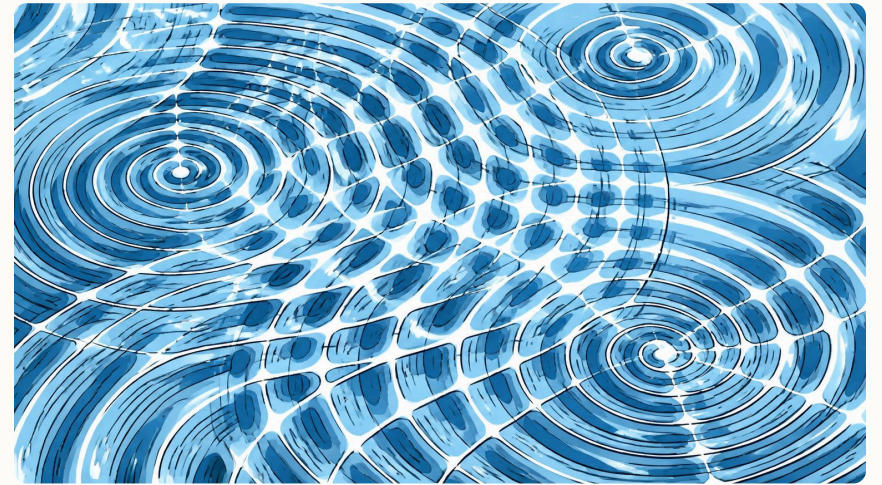
where α and β are complex probability amplitudes satisfying:

$$|\alpha|^2 + |\beta|^2 = 1$$

Example equal superposition state:

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

This enables **quantum parallelism** — many possibilities explored simultaneously.



Measurement of a Qubit

When a qubit in state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is measured, the superposition **collapses** to a definite basis state.



Before Measurement

After Measurement

Measurement is probabilistic and irreversible — once collapsed, the original superposition is lost.



Entanglement

Entanglement is a quantum phenomenon where two or more qubits become **strongly correlated**. A change to one qubit is instantly reflected in the other, even across vast distances.

A canonical example is the **Bell State**:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Strong Correlations

Measurement outcomes are perfectly correlated.

Shared State

Cannot be described as independent qubits.

Non-Local

Correlations persist regardless of distance.

Applications of Entanglement

Entanglement is not just a curiosity — it is the engine behind many quantum technologies.



Quantum Teleportation

Transfer quantum states between distant locations without physical transmission of the particle.



Quantum Networks

Connect quantum processors across distances for distributed quantum computing.



Quantum Cryptography

Enable theoretically unbreakable communication through protocols like BB84 and E91.

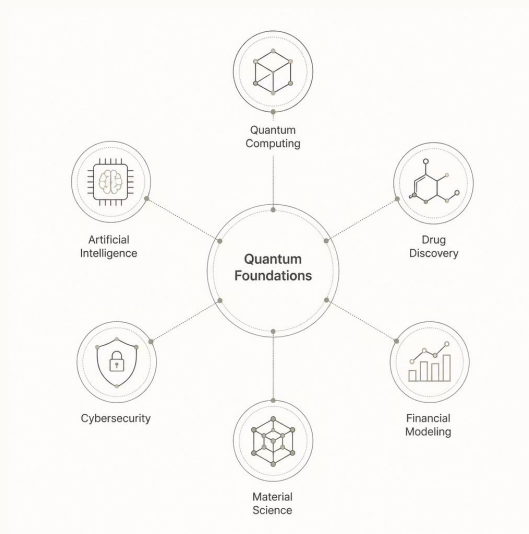


Error Correction

Protect quantum information from decoherence using entangled redundancy.

Why These Concepts Matter

Together, qubits, superposition, and entanglement unlock capabilities far beyond classical computing.



→ Faster Solutions

Exponential speedups for factoring, search, and simulation.

→ Secure Communication

Quantum key distribution guarantees detection of eavesdropping.

→ Advanced Simulations

Model molecules, materials, and physical systems with high fidelity.

Key Takeaways

01

Qubit

The basic unit of quantum information — exists in a combination of $|0\rangle$ and $|1\rangle$.

03

Measurement

Collapses the qubit to a definite state with probability determined by its amplitudes.

02

Superposition

A qubit can occupy multiple states simultaneously, described by complex amplitudes.

04

Entanglement

Creates strong, non-local correlations between qubits — a uniquely quantum resource.

- ❏ These three concepts — qubits, superposition, and entanglement — form the complete foundation of quantum computation and all quantum algorithms.